

Evaluation results

There are three levels of importance in pitfalls according to their impact on the ontology:

- **Critical** It is crucial to correct the pitfall. Otherwise, it could affect the ontology consistency, reasoning, applicability, etc.
- **Important** Though not critical for ontology function, it is important to correct this type of pitfall.
- **Minor** It is not really a problem, but by correcting it we will make the ontology nicer.

Pitfalls detected:

Results for P04: Creating unconnected ontology elements.

2 cases

Minor

Ontology elements (classes, object properties and datatype properties) are created isolated, with no relation to the rest of the ontology.

- This pitfall appears in the following elements:
 - > <http://xmlns.com/foaf/0.1/Person>
 - > <http://www.w3.org/2004/02/skos/core#Concept>

Results for P08: Missing annotations.

27 cases

Minor

This pitfall consists in creating an ontology element and failing to provide human readable annotations attached to it. Consequently, ontology elements lack annotation properties that label them (e.g. `rdfs:label`, `lemon:LexicalEntry`, `skos:prefLabel` or `skos:altLabel`) or that define them (e.g. `rdfs:comment` or `dc:description`). This pitfall is related to the guidelines provided in [5].

- The following elements have neither `rdfs:label` or `rdfs:comment` (nor `skos:definition`) defined:
 - > <http://www.w3.org/ns/dqv#QualityMeasurement>
 - > <http://xmlns.com/foaf/0.1/Person>
 - > <http://www.w3.org/2004/02/skos/core#Concept>
 - > <http://www.w3.org/ns/dqv#Metric>

- The following elements have neither `rdfs:comment` or `skos:definition` defined:
 - > <http://www.w3.org/ns/prov#Agent>
 - > <http://www.w3.org/ns/prov#Role>
 - > <http://www.w3.org/ns/prov#Collection>
 - > <http://www.w3.org/ns/prov#SoftwareAgent>
 - > <http://www.w3.org/ns/prov#Organization>
 - > <http://www.w3.org/ns/prov#EmptyCollection>
 - > <http://www.w3.org/ns/prov#Entity>
 - > <http://www.w3.org/ns/prov#Activity>
 - > <http://www.w3.org/ns/prov#Location>
 - > <http://www.w3.org/ns/prov#Person>
 - > <http://www.w3.org/ns/prov#generated>
 - > <http://www.w3.org/ns/prov#invalidated>
 - > <http://www.w3.org/ns/prov#influenced>
 - > <http://www.w3.org/ns/prov#wasInvalidatedBy>
 - > <http://www.w3.org/ns/prov#specializationOf>
 - > <http://www.w3.org/ns/prov#entity>
 - > <http://www.w3.org/ns/prov#alternateOf>
 - > <http://www.w3.org/ns/prov#activity>
 - > <http://www.w3.org/ns/prov#wasGeneratedBy>
 - > <http://www.w3.org/ns/prov#hadPrimarySource>
 - > <http://www.w3.org/ns/prov#hadMember>
 - > <http://www.w3.org/ns/prov#agent>
 - > <http://www.w3.org/ns/prov#value>

Results for P11: Missing domain or range in properties.

3 cases

Important

Object and/or datatype properties without domain or range (or none of them) are included in the ontology.

- This pitfall appears in the following elements:

- <http://www.w3.org/ns/prov#influenced>
 - <http://xmlns.com/foaf/0.1/homepage>
 - <http://www.w3.org/ns/prov#value>

- **Tip:** Solving this pitfall may lead to new results for other pitfalls and suggestions. We encourage you to solve all cases when needed and see what else you can get from OOPS!

Results for P13: Inverse relationships not explicitly declared.

47 cases

Minor

This pitfall appears when any relationship (except for those that are defined as symmetric properties using owl:SymmetricProperty) does not have an inverse relationship (owl:inverseOf) defined within the ontology.

- OOPS! has the following suggestions for the relationships without inverse:

- <http://www.w3.org/ns/prov#specializationOf> could be inverse of <http://www.w3.org/ns/prov#alternateOf>
 - <http://www.w3.org/ns/prov#alternateOf> could be inverse of <http://www.w3.org/ns/prov#wasQuotedFrom>
 - <http://www.w3.org/ns/prov#wasQuotedFrom> could be inverse of <http://www.w3.org/ns/prov#hadPrimarySource>
 - <http://www.w3.org/ns/prov#hadPrimarySource> could be inverse of <http://www.w3.org/ns/prov#wasDerivedFrom>

- Sorry, OOPS! has no suggestions for the following relationships without inverse:

- <http://www.w3.org/ns/prov#wasRevisionOf>
 - <http://w3id.org/mmv/0.1#hasInput>
 - <http://w3id.org/mmv/0.1#hasPlannedQualityMetric>
 - <http://w3id.org/mmv/0.1#hasStakeholder>
 - <http://w3id.org/mmv/0.1#wasRevisionOfMapping>
 - <http://w3id.org/mmv/0.1#hasQualityMeasurement>
 - <http://xmlns.com/foaf/0.1/homepage>
 - <http://www.w3.org/ns/prov#qualifiedQuotation>
 - <http://www.w3.org/ns/prov#qualifiedAssociation>
 - <http://www.w3.org/ns/prov#atLocation>
 - <http://www.w3.org/ns/prov#hadRole>
 - <http://www.w3.org/ns/prov#hadGeneration>
 - <http://www.w3.org/ns/prov#hadUsage>
 - <http://www.w3.org/ns/prov#qualifiedGeneration>
 - <http://www.w3.org/ns/prov#wasStartedBy>
 - <http://www.w3.org/ns/prov#qualifiedPrimarySource>
 - <http://www.w3.org/ns/prov#entity>
 - <http://www.w3.org/ns/prov#qualifiedAttribution>
 - <http://www.w3.org/ns/prov#influencer>
 - <http://www.w3.org/ns/prov#activity>
 - <http://www.w3.org/ns/prov#used>
 - <http://www.w3.org/ns/prov#hadPlan>
 - <http://www.w3.org/ns/prov#wasAssociatedWith>
 - <http://www.w3.org/ns/prov#qualifiedDelegation>
 - <http://www.w3.org/ns/prov#wasInformedBy>
 - <http://www.w3.org/ns/prov#qualifiedCommunication>
 - <http://www.w3.org/ns/prov#qualifiedEnd>
 - <http://www.w3.org/ns/prov#wasEndedBy>
 - <http://www.w3.org/ns/prov#qualifiedInvalidation>
 - <http://www.w3.org/ns/prov#hadActivity>
 - <http://www.w3.org/ns/prov#qualifiedInfluence>
 - <http://www.w3.org/ns/prov#qualifiedUsage>
 - <http://www.w3.org/ns/prov#qualifiedStart>
 - <http://www.w3.org/ns/prov#wasAttributedTo>
 - <http://www.w3.org/ns/prov#qualifiedRevision>
 - <http://www.w3.org/ns/prov#actedOnBehalfOf>
 - <http://www.w3.org/ns/prov#hadMember>
 - <http://www.w3.org/ns/prov#agent>
 - <http://www.w3.org/ns/prov#qualifiedDerivation>

Results for P19: Defining multiple domains or ranges in properties.

3 cases

Critical

The domain or range (or both) of a property (relationships and attributes) is defined by stating more than one rdfs:domain or rdfs:range statements. In OWL multiple rdfs:domain or rdfs:range axioms are allowed, but they are interpreted as conjunction, being, therefore, equivalent to the construct owl:intersectionOf. This pitfall is related to the common error that appears when defining domains and ranges described in [7].

- This pitfall appears in the following elements:

- <http://www.w3.org/ns/prov#hadActivity>
 - <http://www.w3.org/ns/prov#hadRole>
 - <http://www.w3.org/ns/prov#wasRevisionOf>

Suggestions or warnings:

SUGGESTION: symmetric or transitive object properties| 10 cases

According to the highest importance level of pitfall found in your ontology the conformace badge suggested is "Critical pitfalls" (see below). You can use the following HTML code to insert the badge within your ontology documentation:



```
<p>
<a href="http://oops.linkeddata.es">
</a>
</p>
```

References

- [1] Aguado-De Cea, G., Montiel-Ponsoda, E., Poveda-Villalón, M., and Giraldo-Pasmin, O.X. (2015).
- [2] Noy, N. F., McGuinness, D. L., et al. (2001).
- [3] Gómez-Pérez, A. (1999).
- [4] Montiel-Ponsoda, E., Vila Suero, D., Villazón-Terrazas, B., Dunsire, G., Escolano Rodríguez, E., Gómez-Pérez, A. (2011).
- [5] Vrandečić, D. (2010).
- [6] Gómez-Pérez, A. (2004).
- [7] Rector, A., Drummond, N., Horridge, M., Rogers, J., Knublauch, H., Stevens, R., Wang, H., and Wroe, C. (2004).
- [8] Hogan, A., Harth, A., Passant, A., Decker, S., and Polleres, A. (2010).
- [9] Archer, P., Goedertier, S., and Loutas, N. (2012).
- [10] Bernes-Lee Tim. (2006).
- [11] Heath, T. and Bizer, C. (2011).
- [12] Vatan, B. (2012).

Enter your ontology to scan:

```
<?xml version="1.0"?>
<rdf:RDF xmlns="http://w3id.org/mmv/0.1#"
xml:base="http://w3id.org/mmv/0.1"
xmlns:dqv="http://www.w3.org/ns/dqv#"
xmlns:mmv="http://w3id.org/mmv/0.1#"
xmlns:mod="http://www.isibang.ac.in/ns/mod#"
xmlns:mqv="https://alex-randles.github.io/MQV/#"
xmlns:owl="http://www.w3.org/2002/07/owl#">
```

☐ Uncheck this checkbox if you don't want us to keep a copy of your ontology.

How to cite OOPS!

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